

A zero-sum selection game on matrices: dimension, the order-three case, and complexity

Vamshi Jandhyala

Abstract

Two players play on a real $n \times n$ matrix. The Enemy selects i entries of row i for each i ; then You choose one selected entry in every row, and You win if the chosen entries sum to zero. V. Lev asked on MathOverflow for a description of the matrices on which You win. We show that the winning matrices form a finite union of rational linear subspaces of dimension exactly $n(n+1)/2 - 1$, with the analogous statement for arbitrary row lengths and quotas; the proof is a short rank argument on zero-sum transversals. For $n = 3$ we give a complete description, including five explicit families when every row has distinct entries. For general n , deciding whether a given integer matrix is winning is Π_2^P -complete, already when each row takes at most two values. So a description that can be checked in polynomial time does not exist unless the polynomial hierarchy collapses. The rank argument, the order-three classification and the two-valued normal form behind the hardness proof are formally verified in Lean.

1 Introduction

Here is the game, in the form Lev posed it [2]. Two players, You and the Enemy, look at a real $n \times n$ matrix. First the Enemy selects one entry of the first row, two entries of the second row, and so on, down to all n entries of the last row. Then You choose one selected entry in each row. You win if your n entries sum to 0. Which matrices are winning for You? Lev notes that the question arose in research, not as a puzzle; it is a question about which multisets of numbers force a zero sum against an adversary who removes options row by row.

For $n = 2$ the answer is quick. The Enemy offers one entry a of the first row and both entries c, d of the second, so You need $-a \in \{c, d\}$ for each of the two possible a . Up to the order within rows the winning matrices are

$$\begin{pmatrix} a & b \\ -a & -b \end{pmatrix}, \quad \begin{pmatrix} a & a \\ -a & c \end{pmatrix}.$$

For $n = 3$ the first row is a single forced entry a , the third row is fully available, and all the play is in the middle row, where the Enemy hides one of three entries. Figure 1 shows a winning matrix with distinct entries in every row:

$$M = \begin{pmatrix} -1 & -2 & -3 \\ 0 & 1 & 2 \\ 0 & 1 & 3 \end{pmatrix}.$$

Whatever the Enemy does, You can reply: for $a = -1$ the second-row entries 0 and 1 both complete to zero ($-1 + 0 + 1 = 0$, $-1 + 1 + 0 = 0$), so hiding one of them does not help the Enemy, and similarly $a = -2$ is served by 1, 2 and $a = -3$ by 0, 2. The criterion is exactly this: every first-row entry must have two usable middle entries (Lemma 7).

Lev asked for a description of all winning matrices. The question has been open on MathOverflow since 2023; the comments record the case $n = 2$, examples for $n = 3$ and $n = 4$, and the observation that only the multiset of entries in each row matters [2]. We prove three things.

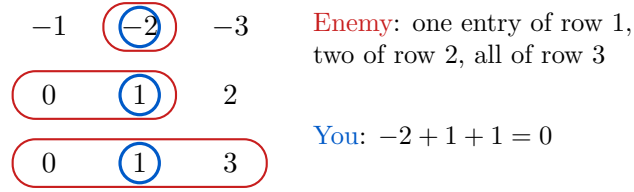


Figure 1: One round on the matrix M . The Enemy's selection is boxed; Your zero-sum reply is circled.

Theorem 1. *The set of winning real $n \times n$ matrices is a finite union of linear subspaces of $\mathbb{R}^{n \times n}$ defined over $\mathbb{Q}$, and its dimension is exactly $\binom{n+1}{2} - 1$. More generally, if row i has N_i entries and the Enemy selects k_i of them ($1 \leq k_i \leq N_i$), the winning set has dimension exactly $\sum_i k_i - 1$.*

Theorem 2. *For $n = 3$, a matrix is winning if and only if every first-row entry a has at least two second-row positions j with $-a - b_j$ in the third row. If every row has distinct entries, the winning matrices are, up to the order within rows and the translation $(A, B, C) \mapsto (A+u+v, B-u, C-v)$, exactly the five families of Table 2.*

Theorem 3. *Deciding whether a given integer matrix is winning is Π_2^P -complete, even when every row has at most two distinct entries.*

Theorem 1 says that winning has codimension exactly $\binom{n}{2} + 1$, and Theorem 3 says that no simple test separates the pieces. The order-three answer, Theorem 2, shows what the pieces look like when they can be listed.

2 Zero transversals

A *transversal* t picks a position t_i in each row; it is a *zero transversal* of the matrix a if $\sum_i a_{i,t_i} = 0$. The Enemy's move is a product of position sets $S_1 \times \cdots \times S_n$ with $|S_i| = k_i$, a *box*, and You win against it when some zero transversal lies in the box. So a is winning exactly when its zero transversals meet every box.

It is convenient to turn the Enemy's selection around. Selecting k_i of N_i positions is the same as deleting the other $q_i = N_i - k_i$. So a is winning if and only if every way of deleting at most q_i positions from row i , for all i , leaves a zero transversal untouched.

For a transversal t let $v_t \in \{0, 1\}^N$, $N = \sum_i N_i$, be its incidence vector, with a 1 in each chosen position. Then t is a zero transversal of a exactly when $v_t \cdot a = 0$. Whether a wins depends only on its set H of zero transversals, and every b with $v_t \cdot b = 0$ for all $t \in H$ has at least those zero transversals. Hence the winning set is the union, over the finitely many sets H of transversals that meet every box, of the subspaces $H^\perp = \{b : v_t \cdot b = 0 \text{ } (t \in H)\}$. These are defined by equations with coefficients 0 and 1, so they are defined over $\mathbb{Q}$. This proves the first assertion of Theorem 1; the content is the dimension.

3 The dimension

Lemma 4 (rank lemma). *If a set H of transversals meets every box, then the vectors v_t , $t \in H$, span a space of dimension at least $1 + \sum_i (N_i - k_i)$.*

The idea is to delete positions one at a time, always a position used by a transversal that has survived so far, and to watch the span shrink (Figure 2).

Proof. Write $q_i = N_i - k_i$ and $q = \sum_i q_i$. By the reformulation above, whatever set of at most q_i positions is deleted from each row, some transversal of H avoids all deleted positions. Fix

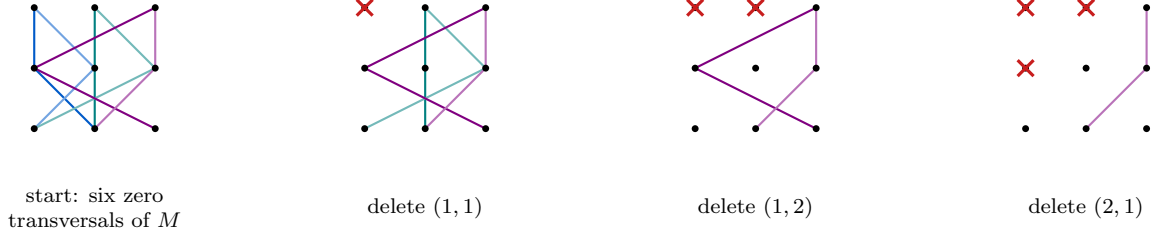


Figure 2: The deletion argument on the winning matrix M of Figure 1. Rows are drawn horizontally; a line through one position per row is a zero transversal. The quotas 1, 2, 3 allow two deletions in row 1 and one in row 2. Each deletion removes a position used by a surviving transversal, which removes that transversal's incidence vector from the span of the survivors, so the span drops in dimension. After all three deletions a transversal still survives, as it must, so the six vectors span at least 4 dimensions.

any order of q deletions that deletes q_i times from row i . Before each deletion, pick a surviving transversal t and delete its position in the scheduled row. That position was not deleted before, since t survived. The surviving transversals now all have a 0 in that coordinate, while v_t has a 1 there, so the span of the surviving incidence vectors lies in a hyperplane that does not contain v_t , and its dimension drops by at least one. After q deletions a transversal still survives, so the final span has dimension at least 1. Hence the initial span has dimension at least $q + 1$. $\square$

Proof of Theorem 1. For each winning a the subspace $H(a)^\perp$ through a has dimension at most $N - (1 + \sum_i (N_i - k_i)) = \sum_i k_i - 1$ by Lemma 4. For the lower bound, put a value b_i in $N_i - k_i + 1$ positions of row i and arbitrary values in the other $k_i - 1$ positions, with $\sum_i b_i = 0$. Any k_i positions of row i include one carrying b_i , so choosing those wins. The parameters are the $\sum_i (k_i - 1)$ free entries and the b_i subject to one equation, $\sum_i k_i - 1$ in all, and they enter injectively. For the square game $\sum_i k_i = 1 + 2 + \dots + n$. $\square$

The same argument bounds arithmetic rather than geometric freedom: for a winning matrix the entries span a $\mathbb{Q}$ -vector space of dimension at most $\sum_i k_i - 1$, because the $\mathbb{Q}$ -linear map $c \mapsto c \cdot a$ on $\mathbb{Q}^N$ has at least $1 + \sum_i (N_i - k_i)$ independent incidence vectors in its kernel. Also, any linear (or affine) family of winning matrices lies inside a single subspace $H^\perp$, because a real vector space is not a finite union of proper subspaces; so the bound applies to every such family.

Remark 5 (distinct entries). The extremal family above repeats values. For even $n = 2r$ the bound is also attained with distinct entries in every row. Take arbitrary rows $x_1, \dots, x_r$ and constants $c_1, \dots, c_r$ with $\sum c_i = 0$, and let row $n + 1 - i$ be $c_i \mathbf{1} - x_i$ (entries in any order). The Enemy selects i positions of row i and $n + 1 - i$ positions of row $n + 1 - i$; under the matching of positions these two sets meet, since $i + (n + 1 - i) > n$. Choosing a common position in each pair gives the sum $\sum_i c_i = 0$. The family has $rn + r - 1 = \binom{n+1}{2} - 1$ parameters, and generic choices give distinct entries. We do not know the answer for odd $n \geq 3$ with distinct entries; for $n = 3$ the maximum is 4 (Section 4).

Remark 6 (a family from Hall's theorem). Let row i be $(r_i + c_1, \dots, r_i + c_n)$, in any order, with $\sum_i r_i + \sum_j c_j = 0$. Label the positions of every row by $1, \dots, n$ through the c_j . The Enemy's selected label sets have sizes $1, 2, \dots, n$, so any m of them have a union of size at least m ; by Hall's theorem they have a system of distinct representatives, which uses every label once and sums to $\sum_i r_i + \sum_j c_j = 0$. This family has dimension $2n - 2$ and, for $n = 3$, is the generic member of the two-parameter family I below.

4 Order three

Lemma 7. *A 3×3 matrix with rows A, B, C is winning if and only if, for every entry a of A , at least two positions j of B have $-a - b_j \in C$.*

Proof. The Enemy offers one entry a of A , two positions of B and all of C . You can reply exactly when one of the two offered positions j has $-a - b_j \in C$. Every pair of positions contains a usable one if and only if at most one position is unusable. $\square$

When the second row has a repeated value b , the criterion reads $A \subseteq -b - C$, since the two copies of b are usable together or not at all. The interesting case is three distinct second-row values, and Theorem 2's five families come from it. Translating B down by $u = \min B$, C down by $v = \min C$ and A up by $u + v$ preserves every sum $a + b + c$, so we may take

$$B = (0, p, p + q), \quad C = (0, r, r + s), \quad p, q, r, s > 0,$$

when B and C have distinct entries. Then a is usable at two positions exactly when $-a$ is a sum $x + y$ with $x \in B$, $y \in C$ in two different ways (two ways with the same x would need the same y). Two such representations $x + y = x' + y'$ say that a difference of B equals a difference of C .

Lemma 8 (collisions). *If t has two representations $t = x + y$, then t and the gaps satisfy one of the nine rows of Table 1. Conversely each row gives two representations of its value.*

B -difference	C -difference	doubled value t
p	r or $r + s$	p
p	s	$p + r$
q or $p + q$	r or $r + s$	$p + q$
q or $p + q$	s	$p + q + r$

Table 1: The nine coincidences of a difference of $B = (0, p, p + q)$ with a difference of $C = (0, r, r + s)$, and the value $t = x + y = x' + y'$ they double.

Proof. Check the 3×3 choices of (x, y) against (x', y') : a coincidence $x - x' = y' - y > 0$ pairs one of the three positive differences $p, q, p + q$ of B with one of $r, s, r + s$ of C , and in each case t is read off as listed. $\square$

Family	A (up to order)	B	C
I	$(-p, -p - q, -2p - q)$	$(0, p, p + q)$	$(0, p, p + q)$
II	$(-h, -2h, -3h)$	$(0, h, 2h)$	$(0, h, 3h)$
III	$(-2h, -3h, -4h)$	$(0, h, 2h)$	$(0, 2h, 3h)$
IV	$(-h, -2h, -3h)$	$(0, h, 3h)$	$(0, h, 2h)$
V	$(-2h, -3h, -4h)$	$(0, 2h, 3h)$	$(0, h, 2h)$

Table 2: The winning 3×3 matrices with distinct entries in every row, normalised so that $\min B = \min C = 0$; $p, q, h > 0$. The matrix of Figure 1 is family II with $h = 1$.

Proof of Theorem 2, distinct entries. The four possible doubled values are $p, p + r, p + q, p + q + r$, and a winning matrix with distinct first-row entries needs three distinct ones. Suppose, for instance, that $p, p + r$ and $p + q$ are all doubled. Then $p = s$ (the only way to double $p + r$),

so $p = r + s$ is impossible and $p = r$; with $r = s = p$ the conditions for $p + q$ become $q = p$ or $q = 2p$, which is family I with $q = p$ or family IV. The other three choices of three values are handled in the same way and give families I, II, III and V. Substituting each family's gaps into Table 1 shows that exactly three values are doubled, the negatives of the listed A , so the first row must be that triple. Conversely, Lemma 8 shows that each listed value is doubled, so each family wins. $\square$

Without the distinctness assumption the same method gives a complete list. Up to the order within rows and the translation, every winning 3×3 matrix belongs to one of twelve linear families of dimensions 5, 4, 3; as subspaces of $\mathbb{R}^9$ there are 1107 maximal winning subspaces, 99 of dimension 5, 576 of dimension 4 and 432 of dimension 3. These numbers come from enumerating the minimal sets of zero transversals that meet every box, and two independent programs agree on them (Section 7). The maximum $5 = \binom{4}{2} - 1$ is Theorem 1; with distinct entries in every row the maximum is 4, attained by family I with the translation.

5 Complexity

For larger n there is no comparably short list, and the reason is computational. The key case is a matrix whose rows take at most two values.

Proposition 9. *Suppose every entry of row i is u_i or $w_i \neq u_i$, with α_i and β_i copies. Call row i universal if $\alpha_i, \beta_i \geq k_i$, existential if $\alpha_i, \beta_i < k_i$, and fixed otherwise, with fixed value the one having at least k_i copies. A constant row counts as fixed. Then the matrix is winning if and only if for every choice of u_i or w_i in the universal rows there is a choice in the existential rows such that these values and the fixed values sum to zero.*

Proof. In a universal row the Enemy can select k_i copies of either value, so he decides which value You get; in a fixed row he can always force the fixed value and can never exclude it; in an existential row any k_i positions include both values. A selection of any other kind only offers You more, so the Enemy's best moves are exactly the choices of a value in each universal row, after which You choose freely in the existential rows. $\square$

Proof of Theorem 3. Membership in Π_2^P is the definition of the game: for all Enemy selections there exists a reply, and a reply is checked in polynomial time. For hardness we reduce from GENERALIZED SUBSET SUM: given integers $u_1, \dots, u_p, w_1, \dots, w_q$ and T , decide whether for every $x \in \{0, 1\}^p$ there is $y \in \{0, 1\}^q$ with $\sum u_j x_j + \sum w_j y_j = T$. Its complement is Σ_2^P -complete [1, 4], so it is Π_2^P -complete. We may drop indices with $u_j = 0$ or $w_j = 0$. Let $m = \max(p + 1, q)$ and $n = 2m$. Row 1 is constant $-T$. Rows $2, \dots, p + 1$ consist of m zeros and m copies of $u_1, \dots, u_p$ respectively; since their quotas are at most m , they are universal. Rows $m + 1, \dots, m + q$ consist of m zeros and m copies of $w_1, \dots, w_q$; their quotas exceed m , so they are existential. All other rows are zero, hence fixed. By Proposition 9 the matrix is winning exactly when the subset-sum instance is true, and its size is polynomial in that of the instance. $\square$

So recognising winning matrices is as hard as any problem at the second level of the polynomial hierarchy, and a polynomial-time recognisable description would give $P = \Pi_2^P$. This does not exclude a structural description that is expensive to test, such as the finite union of Section 2, which can be generated from n alone by exact linear algebra; it only rules out a cheap one.

6 Questions

Question. For odd $n \geq 3$, what is the largest dimension of a family of winning matrices with distinct entries in every row? For $n = 3$ it is 4, one less than the unrestricted maximum.

Question. How many maximal winning subspaces are there for $n = 4$? For $n = 3$ there are 1107, in twelve orbits under permutations within rows.

7 Verification

Lemma 4 and Theorem 1 (both bounds, for arbitrary row lengths and quotas, and the finite-union statement), Lemma 7, Lemma 8 and its converse, the distinct-entry classification in normalised form, and Proposition 9 are formally verified in Lean 4 with Mathlib [3], using only the standard axioms. Not formalised are the bookkeeping of the hardness reduction (which rows are universal), the twelve families and the count 1107, and the two remarks of Section 3. A program sharing no code with the formal proofs or with the programs that found these results checks, by playing the game literally: the criterion on all 592,704 matrices with sorted rows from $\{-3, \dots, 3\}$; the five families against all matrices with distinct rows from $\{-4, \dots, 4\}$; the count 1107 by its own enumeration; Proposition 9 on random two-valued matrices; the reduction on random subset-sum instances, true and false; and the families of the remarks. The code is at <https://github.com/jvvk/mathematics>.

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